

RAGASUDHA S.S

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Ragasudha S.S - Portfolio

CAREER OBJECTIVE

Fresher AI Engineer with hands-on experience building production-grade Computer Vision, NLP, and Multi-modal AI systems — including a YOLOv8 defect detection pipeline at 92.6% accuracy and a full-stack RAG-powered web application. Proven ability to ship end-to-end AI solutions using Python, PyTorch, TensorFlow, Flask, and Hugging Face. Interned at Infosys Springboard and led a 60+ member technical club as Secretary. Passionate about building real-world GenAI and ML products that scale.

EDUCATION

- **B.Tech in Artificial Intelligence and Data Science** 2022 – 2026
Kangeyam Institute of Technology, Tirupur CGPA: 8.48
- **HSC – MN. Murugappa Chettiar Girls HSS, Tirupur** 2022 — 85.5%
- **SSLC – Thirumurugan Matric HSS, Tirupur** 2020 — 83.6%

SKILLS

- **Programming Languages:** Python, SQL
- **Machine Learning & Deep Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, XGBoost, Random Forest, Transfer Learning, Model Optimization, Hyperparameter Tuning
- **Computer Vision:** OpenCV, YOLOv8, YOLO Object Detection, Image Classification, Real-time Detection
- **NLP & Generative AI:** Transformers, Hugging Face, TF-IDF, Sentiment Analysis, RAG, CLIP, Vision-Language Models, Prompt Engineering, Multi-modal AI
- **Data Analysis & Visualization:** Power BI, DAX, Tableau, Excel, Power Query, Pandas, NumPy, Matplotlib, Seaborn
- **Databases & Backend:** MySQL, PostgreSQL, SQLite, Flask, FastAPI, Bootstrap
- **Tools & Platforms:** Git, GitHub, KNIME, Jupyter Notebook, Google Colab, VS Code

PROJECTS

BridgeGuard AI: Intelligent Bridge Inspection System Python, YOLOv8, Flask, OpenCV, SQLite

- Engineered a YOLOv8-based real-time object detection system classifying 4 bridge defect types (cracks, corrosion, spalling, rebar exposure) achieving **92.6% mAP accuracy** on a custom-labelled dataset of 3,000+ images.
- Integrated ESP32-CAM and drone feeds with a live Flask dashboard at **~25 fps**; automated PDF inspection reports with geospatial heatmaps and severity scoring.
- Deployed on **Hugging Face Spaces** for public access and demonstration.

Multi-Modal AI Analysis System Python, CLIP, PyTorch, Transformers

- Architected a cross-modal deep learning pipeline jointly processing image and text using CLIP embeddings; implemented cross-modal attention mechanisms improving classification **F1-score by 11%** over single-modality baseline.
- Designed modular encoder architecture allowing independent swapping of vision or language encoders without retraining the fusion layer.

AI Mental Health Assistant Python, Flask, NLP, Transformers, Sentiment Analysis

- Built a conversational AI detecting user mood across **7 emotional states** using a fine-tuned sentiment classifier (**88% accuracy**); routed users to context-appropriate responses via Transformer-based intent recognition.

Fake News Detection System

Python, NLP, TF-IDF, Logistic Regression, Scikit-learn

- Built a binary text classifier using TF-IDF and Logistic Regression achieving **96.4% accuracy**; outperformed Naive Bayes and SVM baselines by 3–4% F1-score with 5× faster inference.

Analytics Dashboards

Power BI, Tableau, Excel, DAX, Power Query

- **Hotel Booking (Power BI/DAX):** Executive dashboard tracking 12-month revenue and occupancy; DAX measures surfaced a 17% off-peak revenue gap, informing promotional strategy.
- **Student Performance (Excel/Power Query):** Automated ingestion of 500-row assessment data; dynamic KPI slicers for subject-wise faculty drill-down.
- **Mental Health in Tech (Tableau):** Visualized 1,200-respondent survey with KPI cards for burnout rate, anxiety prevalence, and work-life balance segmented by company size.

INTERNSHIP EXPERIENCE

Artificial Intelligence Intern – Infosys Springboard

Nov 2025 – Jan 2026

- Developed **EcoPackAI**, a full-stack AI-powered sustainable packaging recommendation system, owning the project end-to-end from data preprocessing and model training to web deployment.
- Trained and benchmarked **Random Forest** and **XGBoost** models; XGBoost achieved **94.2% precision** on held-out test data, outperforming Random Forest by 2.8% F1-score.
- Built a **Flask** REST API and **Bootstrap** front-end integrated with a **PostgreSQL** database for product data storage, recommendation history, and analytics.

Data Analytics Intern – KNIME (Inplant Training)

Aug 2024

- Automated **6 end-to-end data processing workflows** in KNIME, replacing manual Excel pipelines and reducing weekly reporting time by approximately **60%**.
- Integrated KNIME workflow outputs with Excel dashboard templates, enabling non-technical stakeholders to refresh and interpret analytical reports independently.

CERTIFICATIONS

- AI & MLOps – Udemy
- Artificial Intelligence – Infosys Springboard
- Prompt Engineering – Infosys Springboard
- Introduction to Python – Infosys Springboard
- Advanced Computer Networks – NPTEL

ACHIEVEMENTS AND PARTICIPATION

- **Smart India Hackathon (National Level)** – Qualified institutional screening; built and presented an AI-powered solution at the national round.
- **2nd Prize** – Poster Design Competition: “Enhanced AI Technology for Future” (college-level technical event).
- **AI-Driven Hackathon** – Built and demonstrated a functional AI prototype within a 24-hour competitive environment.

ROLES AND RESPONSIBILITIES

- **Secretary – Cyber Champion Club** *Mar 2025 – 2026*
Coordinated workshops and inter-college competitions for **60+ members**; launched monthly AI knowledge-sharing sessions, increasing active participation by **40%**.
- **Treasurer – Association of AI & DS** *Aug 2025 – 2026*
Managed annual budget allocation and expenditure tracking for events, guest lectures, and association activities.
- **Volunteer – Elantra '25 (Annual Technical Symposium)** *2025*